

Lower Crossed Syndrome and Postural Considerations



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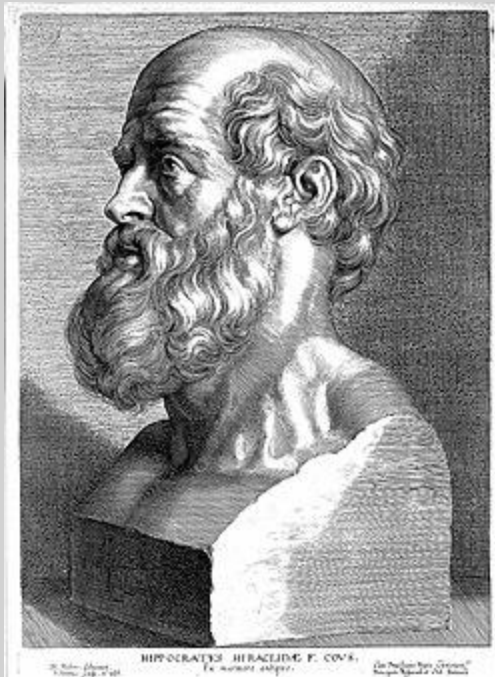
Objectives

1. the students will be able to appreciate the historical background/perspective, from Hippocrates, to A.T. Still, to consideration of the 5 (or 6) Osteopathic Models, to Dr. Phil Greenman's legacy, and on to the related to inter-professional and global awareness of the Lower (and Upper) Crossed syndrome, from the legacy of neurologists, Drs. Janda and Lewit.
2. the students will be able to understand the Lower Crossed Syndrome, including the role of the type I (slow twitch, postural) and type II (fast twitch, phasic) muscles. The student will understand which muscles need stretching, and which muscles need strengthening, in relation to application of an understanding of this "crossed" syndrome. Specific exercises will be reviewed at the end of the lecture, with pictures and diagrams included.
3. the students will be able to understand the importance of obtaining and considering the patient's history (subjective), along with objective information. This objective information can include Special Tests, muscle testing/palpation, the osteopathic structural examination, postural observation and analysis, and other neurological (NCV / EMG) or radiological (x-rays, MRI, etc.) testing, as needed.
4. the students will be able to understand how to incorporate an understanding of postural compensation, tensegrity (a.k.a. biotensegrity), gravitational strain pathophysiology (G.S.P.) and postural decompensation into their osteopathic perspective.
5. the students will read the assigned article from Kim (LINK: <https://pubmed-ncbi-nlm-nih-gov.nyit.idm.oclc.org/30899747/>), which is also listed under the Resources for this Classroom Session) on their own, to gain a sense of the clinical correlation / clinical suggestion related to this topic area.

Outline

- Historical perspective
- Review Muscle Types (I and II)
- Lower (Pelvic) Crossed Syndrome
- Other Postural Considerations
 - Postural Compensation
 - Tensegrity
 - Gravitational Strain Pathophysiology (G.S.P.) and Postural Decompensation
- General Treatment Strategies

Historical Perspective



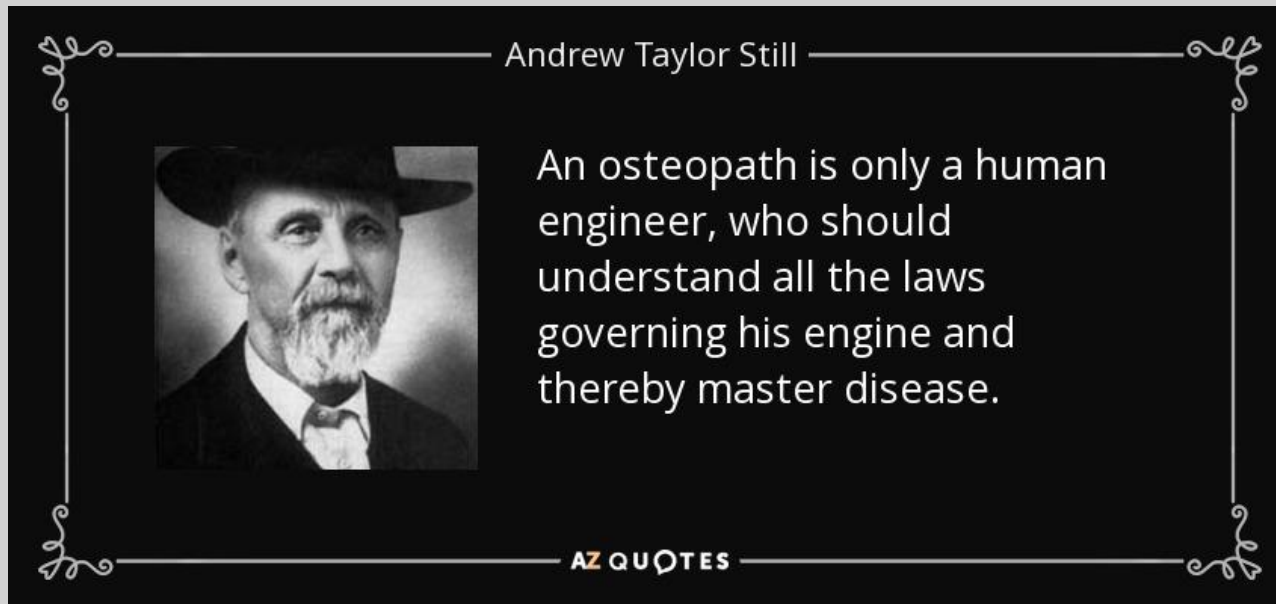
Engraving: bust of Hippocrates by [Paulus Pontius](#) after [Peter Paul Rubens](#), 1638
<https://share.google/3NOJhdAazy25CJg9l>

“Nearly 2,500 years ago, **Hippocrates** said, ‘**The regimen I adopt will be for the benefit of my patients according to my abilities and judgment.**’”

As we learn about Postural Considerations in Osteopathic Diagnosis and Treatment, we can consider:

- “principles intended to enhance the ability to understand postural mechanisms as dynamic processes affecting interrelated patient neuromusculoskeletal function.”
- that “with understanding and practice, the physician’s judgment in prescribing individualized postural treatment regimens will evolve and improve.”

Historical Perspective

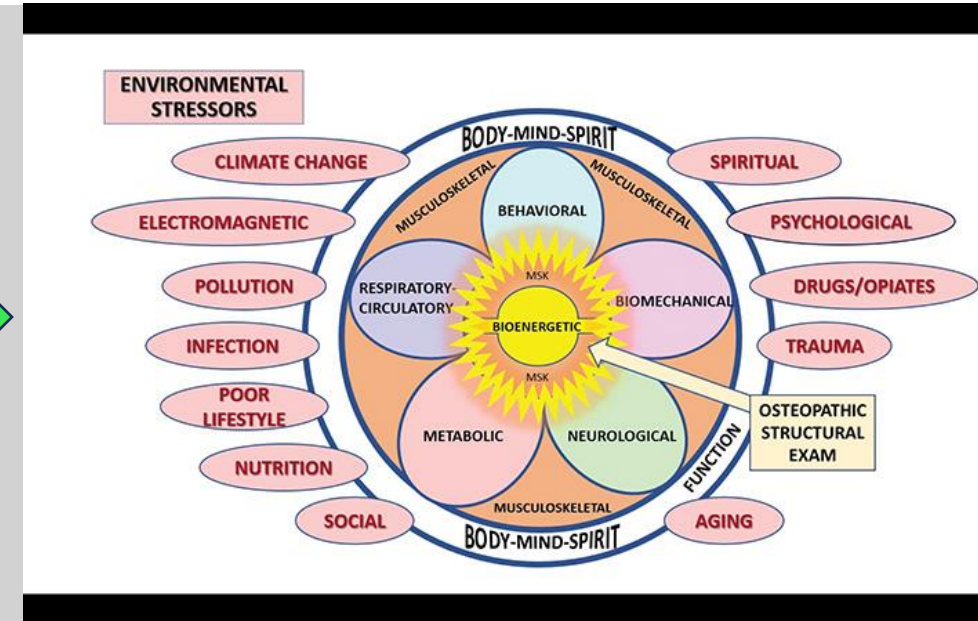
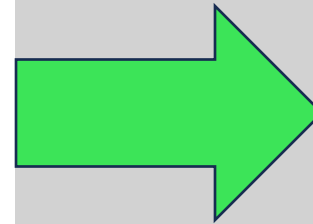
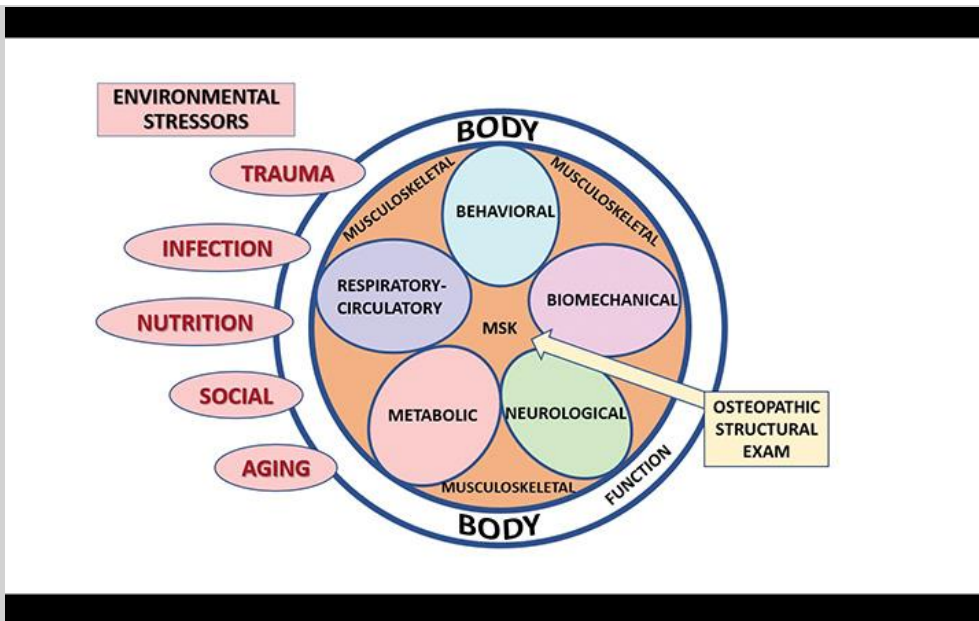


<https://share.google/2j2scczUSyT2AvyUs>
<https://www.lifemedidaho.com/osteopathy>

“Andrew Taylor Still placed much emphasis on **the fascia and its relationship to health**. He wrote, ‘I know of no other part of the body that equals the fascia as a hunting ground (sic. for health/disease) ... By its action we live, and by its failure, we die.’”

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model**.
Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 5e, 2026,
LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456&resultClick=1#261507713>

Historical Perspective



Original five (5) models adopted
by Educational Council on Osteopathic
Principles (ECOP) in 1982.

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model.**
Foundations of Osteopathic Medicine: Philosophy,
Science, Clinical Applications, and Research, 5e, 2026

Department of Osteopathic Manipulative Medicine

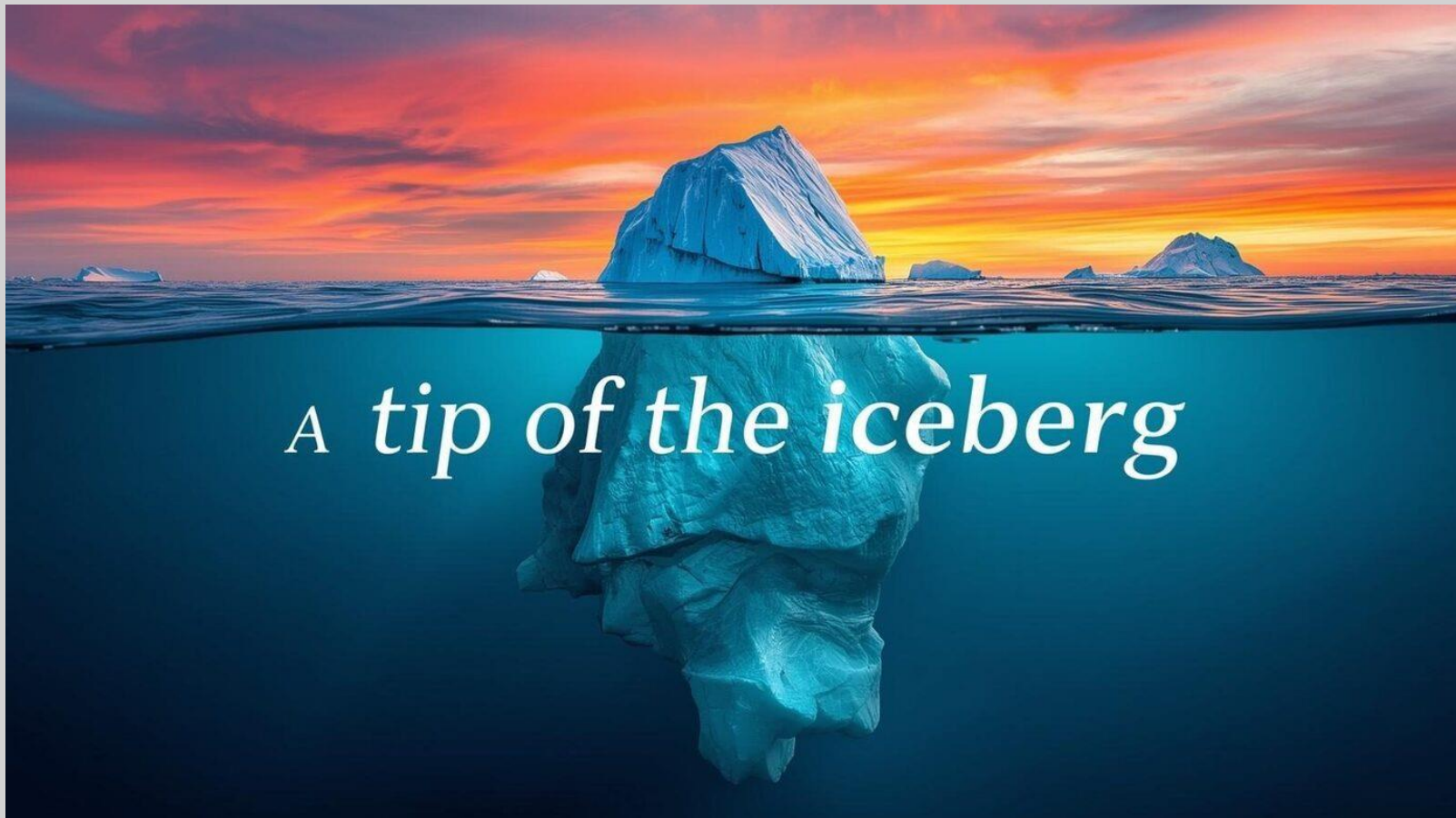
**Expanded six (6) models to
include bioenergetics.**

From: **Chapter 61 Dynamic Strain and the Bioenergetics Model.**
Foundations of Osteopathic Medicine: Philosophy,
Science, Clinical Applications, and Research, 5e, 2026,
LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456&resultClick=1#261507624>

Historical Perspective

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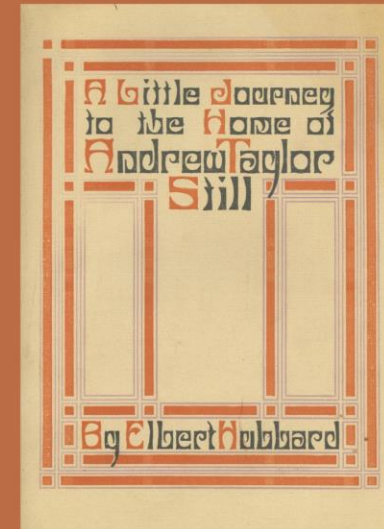
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Historical Perspective

D.O. = “Dig On”
(i.e., continue to
search
and learn!)
in your osteopathic
reasoning!

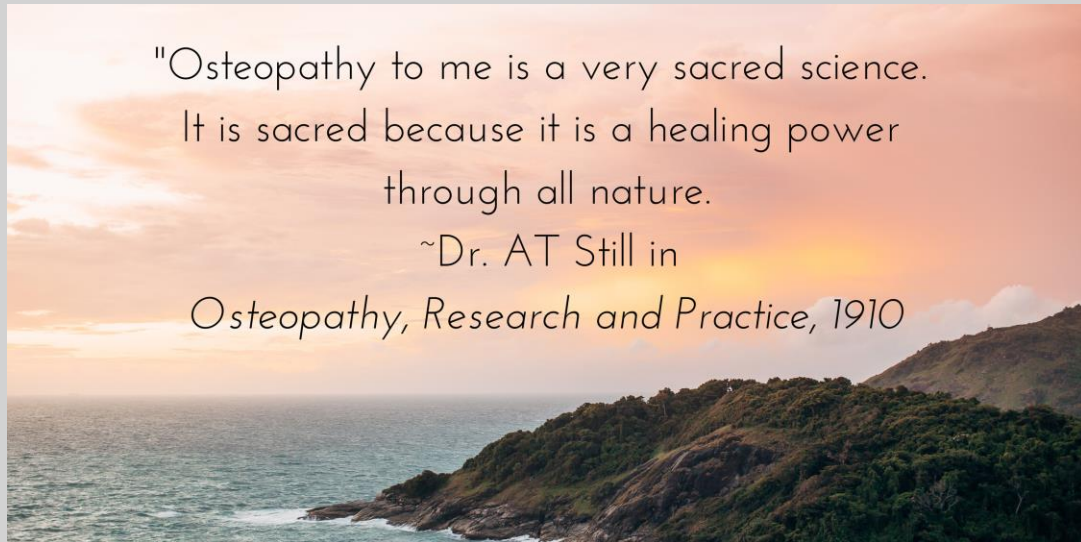
In 1909, the famous publisher and author Elbert Hubbard visited “The Old Doctor” and published this 32-page book which helped to spread the word of Osteopathy.



“Osteopathy is simply the practice of common sense.” ~Elbert Hubbard

Historical Perspective

D.O. = “Dig On” (i.e., continue to search and learn!) in your osteopathic reasoning!



<https://share.google/BzQ8yZdbEBnjVtBTp>

➤ Your “digging on” (your search) may bring you to incorporate information and/or practical approaches, which originate from other fields of reasoning, including but not limited to osteopathic medicine itself; neurology, orthopedics, physiatry, functional medicine, or any other medical specialty or primary care field; bioenergetics, spirituality, metabolism/nutrition, or any of the other components of the “**6 Models**” of Osteopathic Medicine; chemistry, physics (quantum physics), or any of the other “basic” (pre-med) or advanced level sciences; embryology, growth and development, etc.

Historical Perspective

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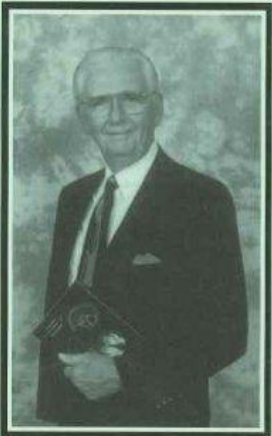
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Philip E. Greenman, DO, FAAO
(1928-2013)

→ “Digging On” in Osteopathic
Medicine throughout his career

GREENMAN'S WORKS

THE COLLECTED WORKS OF
PHILIP E. GREENMAN, DO, FAAO



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AMERICAN ACADEMY OF OSTEOPATHY®
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“What I do know is that when I put my hands on a patient as part of a comprehensive physical examination, I begin to evolve a line of questioning that isn’t entirely conscious. I pick up information in my hands that guide the questions I next need to ask.”

Philip E. Greenman, D.O., FAAO
Michigan State University COM

Dr. Greenman was also aware of the work of Drs. Janda and Lewit and the Crossed Syndromes.

Historical Perspective

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Philip E. Greenman, DO, FAAO
(1928-2013)



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Department of Osteopathic Manipulative Medicine

Philip E. Greenman, DO, FAAO



Dr. Greenman taught osteopathic manipulative medicine at all levels - osteopathic students, interns, residents, fellows, allopathic and osteopathic physicians, and other health care professionals - and achieved international recognition for his teaching in manual medicine tutorials.

His video series is the most complete and comprehensive set of educational materials that is available to practitioners of manual medicine. In these videos, Dr. Greenman introduces the student to basic models of manual medicine as they related to diagnosis and treatment, guided the student through a standard screening examination of a patient, introduced the student to the unique anatomic/biomechanic characteristics of specific regions within the musculoskeletal system, and finally guided the student through the application of comprehensive diagnostic and treatment protocols.

Now deceased, Dr. Greenman served as senior associate dean and professor of osteopathic manipulative medicine and of physical medicine and rehabilitation at the Michigan State University College of Osteopathic Medicine. A charter faculty member at MSUCOM, Dr. Greenman is recognized for both his artistry and scientific knowledge of manual medicine.

"I'd decided that I wanted to be a DO at age eight," Greenman remembered. "Our family doc was an '03 graduate of the Kirksville college, then called the American School of Osteopathy. As a second grader, I saw one of my classmates experience back and leg pain and weakness after a fall. They took her to Albany for treatment, but she didn't get better until she was seen by a DO." From that early decision

<https://share.google/iRSLFYEvqxRhRD8UK>

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)

- H/o poliomyelitis as a teenager, wheelchair-bound for 3 years after an initial quadriplegia.
- **Neurologist** training at Charles University, Prague.
- 1960s in Canada, he did post-doctoral work with Professor John Basmajian.
- His work in rehabilitation included research and clinical activities, over many decades and with a global influence.
- He was the Editor in Chief of the Czech Journal, Rehabilitation and Physical Medicine, until 2002.
- He worked with **Karel Lewit, M.D.**, another Czech **neurologist**, **presenting the Upper & Lower Cross Syndromes.**



Vladimir Janda, MD, DSc 1928–2002.

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)



Vladimir Janda, MD, DSc 1928–2002.

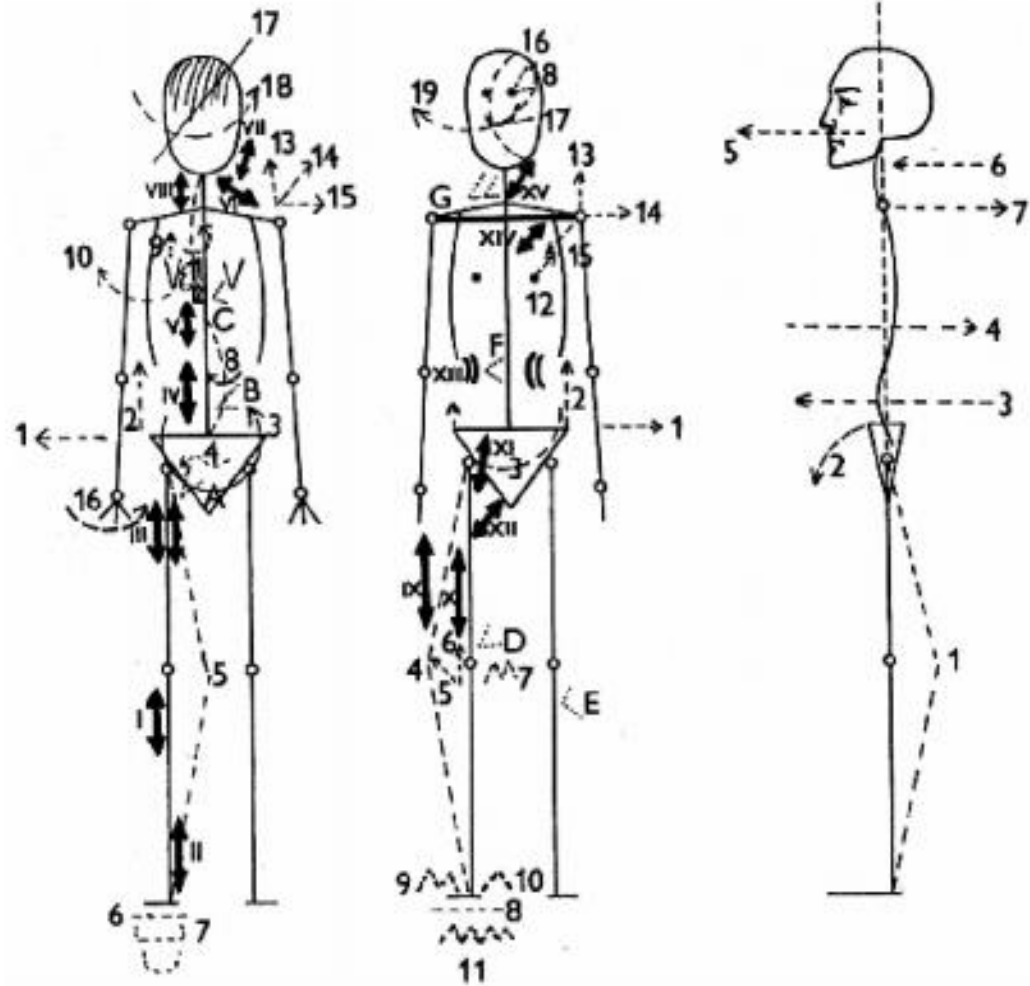
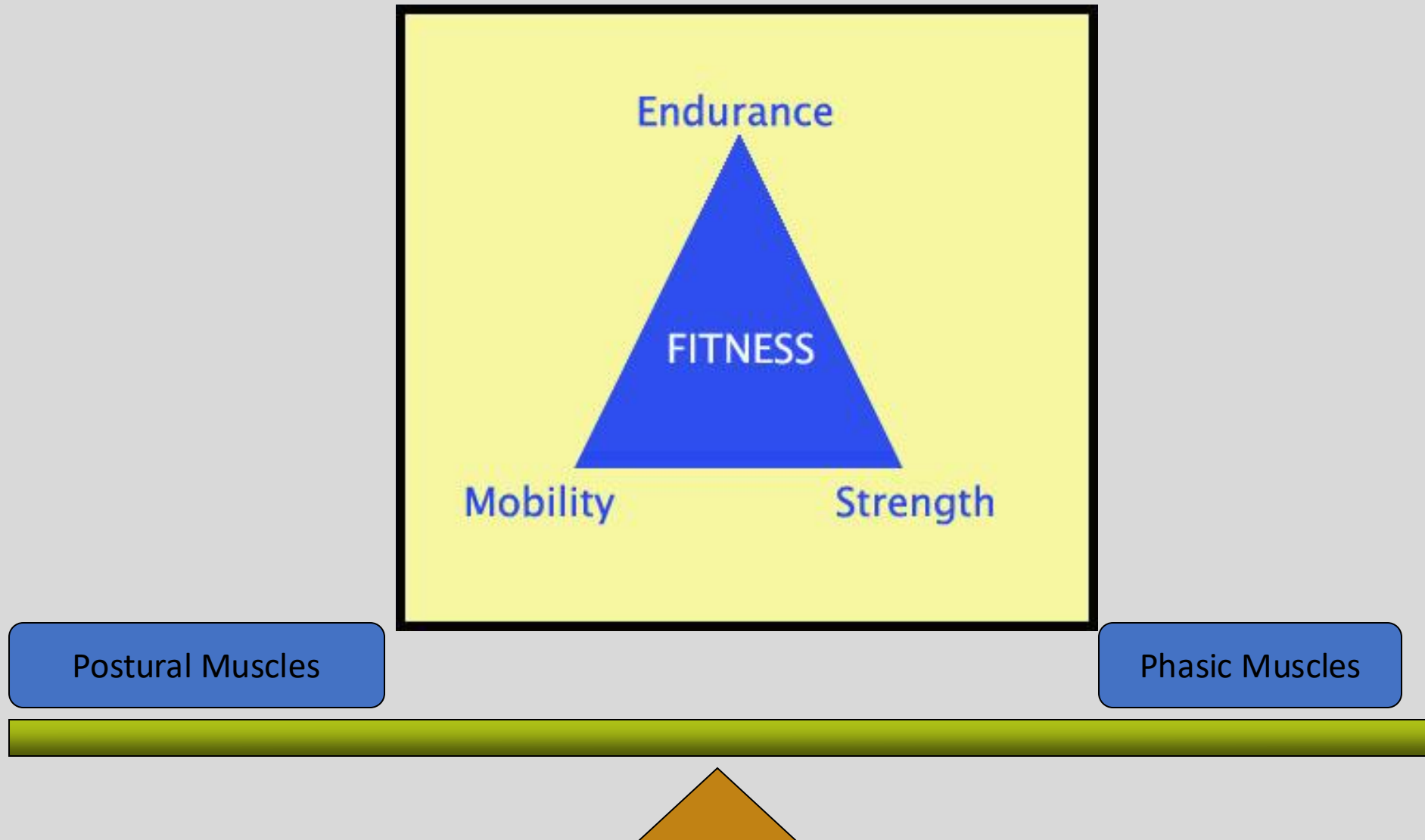


Figure 7. One of Janda's drawings demonstrating postural assessment. He emphasized the local and global assessment as part of a complete examination, teaching methods that allowed the examiner to quickly log his or her findings. (Reproduced from Janda V. Dokumentace analyzy stoje [How to record stance analysis]. *Rehabilitace a Fyzikalni lekarstvi* 1994:4–5, with permission.)

Balance



Which muscles are used?



Ashlyn Wang

The leading runners of the first wave run down Beacon Street in Brookline, Mass., toward the finish line on Monday, April 15, 2024. (Ashlyn Wang/Beacon Staff)
<https://share.google/Oz2tlxzlkbxH5qhX>

- **Type I – tonic, slow-twitch muscles – “long endurance work” – postural muscles – tend to become facilitated and shortened/tight**



Picture by Steph Chambers/WA/Getty Images for World Athletics 2026

<https://www.olympics.com/en/news/tate-taylor-world-u20-champion-one-upping-noah-lyles>

- **Type II – phasic, fast-twitch muscles – fatigue more easily – tend to become inhibited and muscles become weak**

In clinical practice, we distinguish two muscle types in the evaluation and treatment:

**Type I muscles = postural muscles =
slow twitch muscles**

These muscles react by becoming TIGHT or SHORTENED as they become “facilitated” in response to postural imbalance, injury, somatic dysfunction.

Thus it is good to know which muscles are being categorized as postural/slow twitch muscles.

**Type II muscles = phasic muscles =
fast twitch muscles**

These muscles react by becoming WEAK as they become “inhibited,” For instance, the Gluteus medius muscle, a phasic muscle, may become reflexly inhibited due to an associated sacroiliac joint issue (e.g. extended, torsion, etc.) on one side and thus causing the muscle to become relatively weak on that side vs. the other side. The muscle itself has “not lost its innate strength” but rather is inhibited and thus becomes weak as a result.

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
TYPE I	Postural	Slow twitch	Become facilitated and thus SHORTENED or TIGHT	Upper crossed syndrome: Example 1: Head forward posture – <u>Anterior tight/shortened</u>= facilitated muscles: Pectorales <u>Posterior tight /shortened</u>=facilitated muscles: Levator scapulae, upper trapezius, SCM Example 2: when there is excessive ankle pronation – one can see the lower crossed syndrome – <u>Anterior: tight</u> iliopsoas muscle and <u>Posterior: tight</u> erector spinae muscles

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
Type II	Phasic	Fast twitch	Become <u>inhibited</u> and thus WEAK	Upper crossed syndrome (example: head forward posture) <u>Anterior: weak</u> neck flexors=inhibited <u>Posterior: weak</u> rhomboids/serratus anterior LOWER crossed syndrome: <u>Anterior: weak/inhibited</u> abdominals <u>Posterior: weak/inhibited</u> gluteus maximus/medius

Upper Crossed and Lower Crossed Syndromes

Clinically (e.g. work by Vladimir Janda, M.D./Prague/Czech Republic) there have been observed PATTERNS of muscle involvement, that have been called the **upper and lower crossed syndromes**. “Upper” because it involves the upper body, and “lower” because it involves the pelvis, or lower body.

It is called “Crossed” as there are two ‘lines’ that form a cross-over with ANTERIOR muscles being facilitated=tight=shortened (type I muscles) and POSTERIORLY (follow one LINE – call it the “facilitated” or “shortened” line) with type I muscles being also shortened. This is in response to a muscle imbalance.

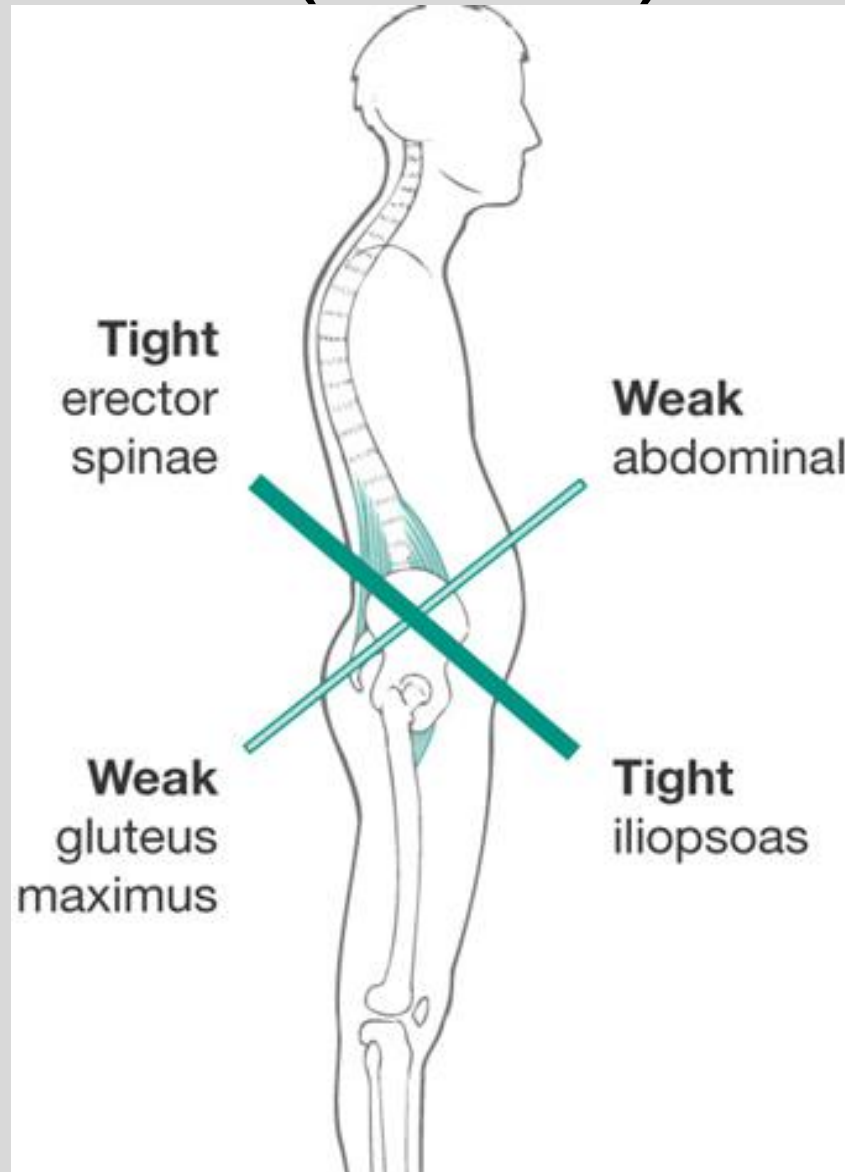
Similarly, follow the other cross-line that goes from ANTERIOR to POSTERIOR, representing the INHIBITED and thus WEAK muscles both anteriorly and posteriorly.

So it is like a “puzzle” where you have two lines with four “stations:” 2 anterior and 2 posterior groups. Knowing the four “groups” – will give you an understanding of which muscles can be involved at any one time – so if the pectoralis major is tight in front, and you do have a cross syndrome – upper, for instance – you can deduce which other muscles are involved in a particular patient.

Lower (Pelvic) Crossed Syndrome

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Lower Crossed Syndrome

A = Tight Line: Observe how tight line (A) traverses the lumbar erector spinae muscles and iliopsoas. Neurologically shortened iliopsoas tissues anteriorly tilt the pelvic bowl creating excessive lumbar lordosis.

B = Weak Line: Connecting the abdominals and gluteals, weak line B permits the lower crossed asymmetry. Core support is lost as the stretch-weakened rectus and transversus abdominal muscles are overpowered by the pull of the strong iliopsoas and erector spinae groups.

Typical Muscle Imbalances in the LOWER CROSSED SYNDROME	
BOX A TIGHT, FACILITATED Iliopsoas Rectus Femoris Hamstrings Erector Spinae Tensor Fascia Lata Thigh Adductors Piriformis Quadratus Lumborum Gastroc/Soleus	BOX B WEAK, INHIBITED Rectus Abdominis Transversus Abdominis Obliques Gluteus Maximus Gluteus Medius/Minimus Vastus Lateralis Vastus Medialis Tibialis Anterior/Posterior Peroneus Longus

POSTURE and The **CROSSED SYNDROMES**

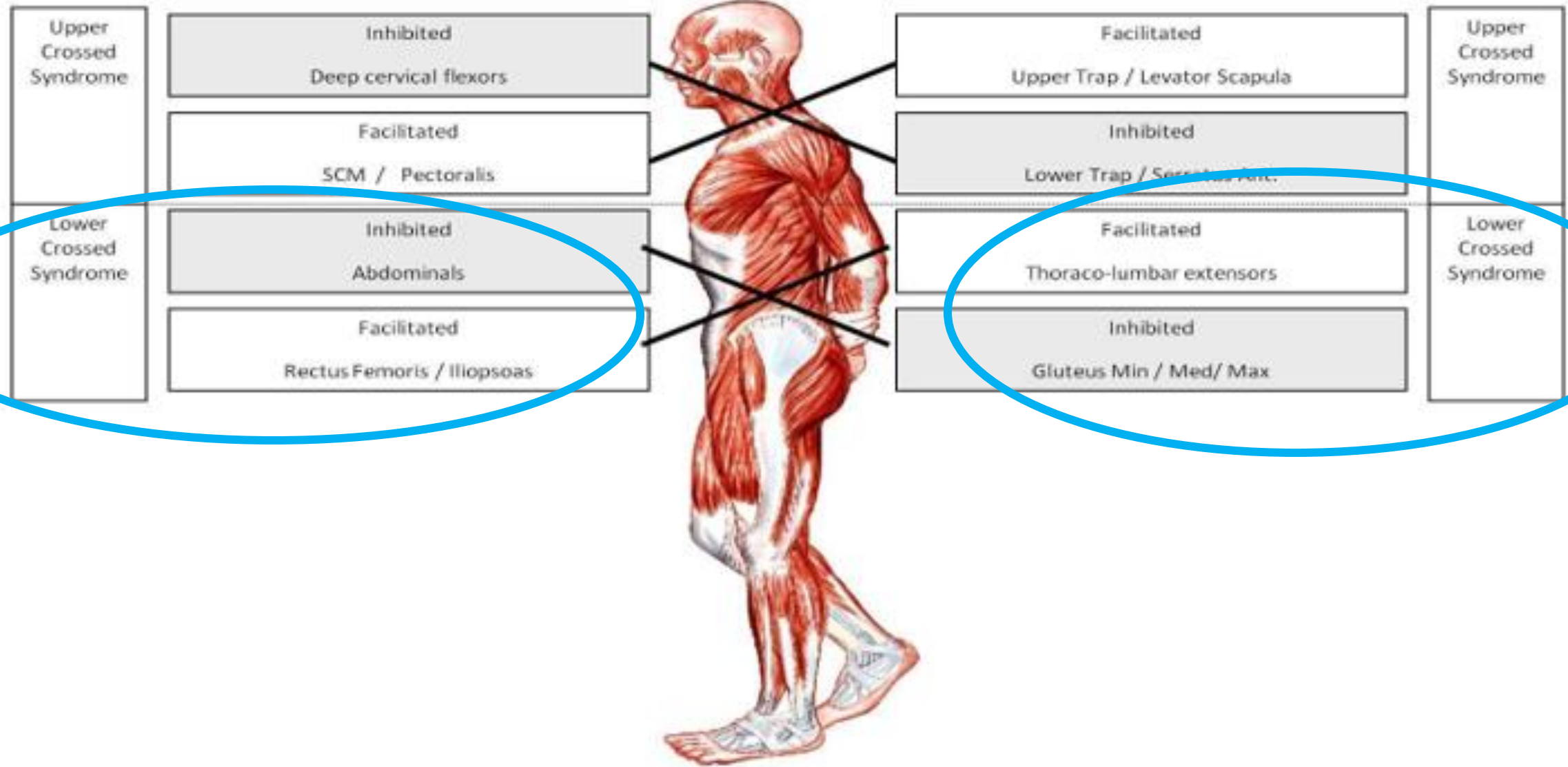


Figure 1 : Janda's Muscle Imbalance Syndromes

Table 1 – Anterior Muscles involved in Upper and Lower Crossed Syndromes

memorize

Anterior Postural Muscles	Anterior Phasic Muscles
Upper Cross	
Scalenes	Deep cervical flexors
Sternocleidomastoid	
Pectoralis major (upper)	
Pectoralis minor	
Lower Cross	
Iliopsoas	Rectus abdomini
Rectus femorus	Three Vasti muscles
Adductor brevis	Tibialis anterior

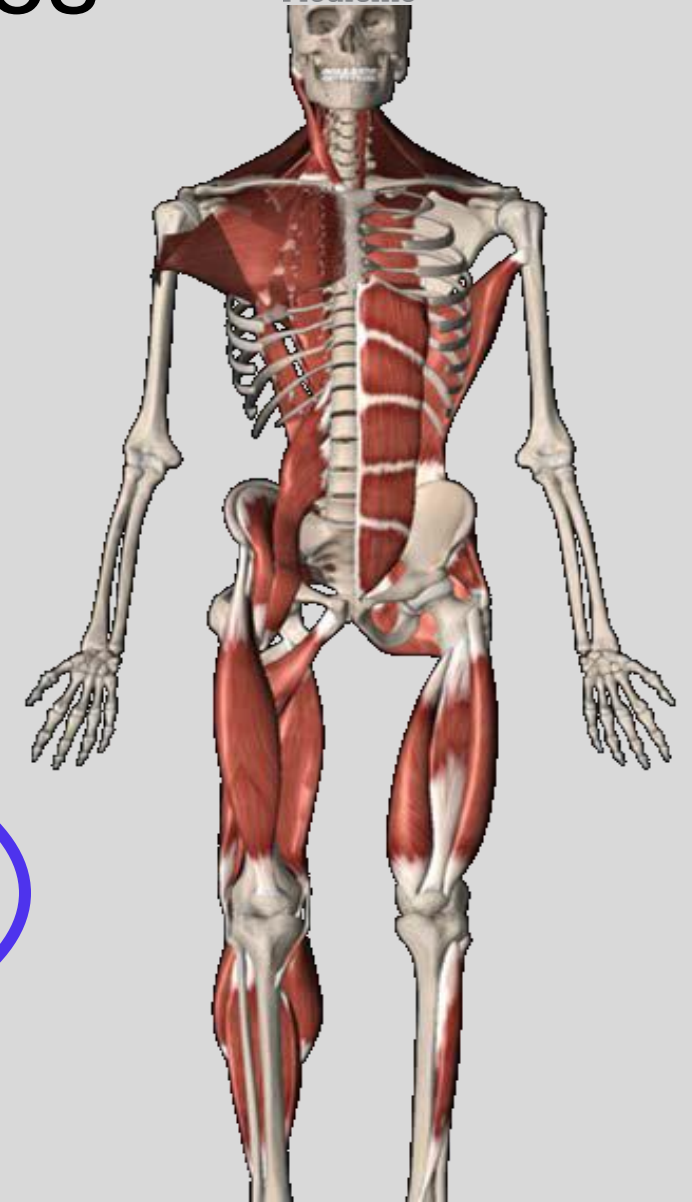
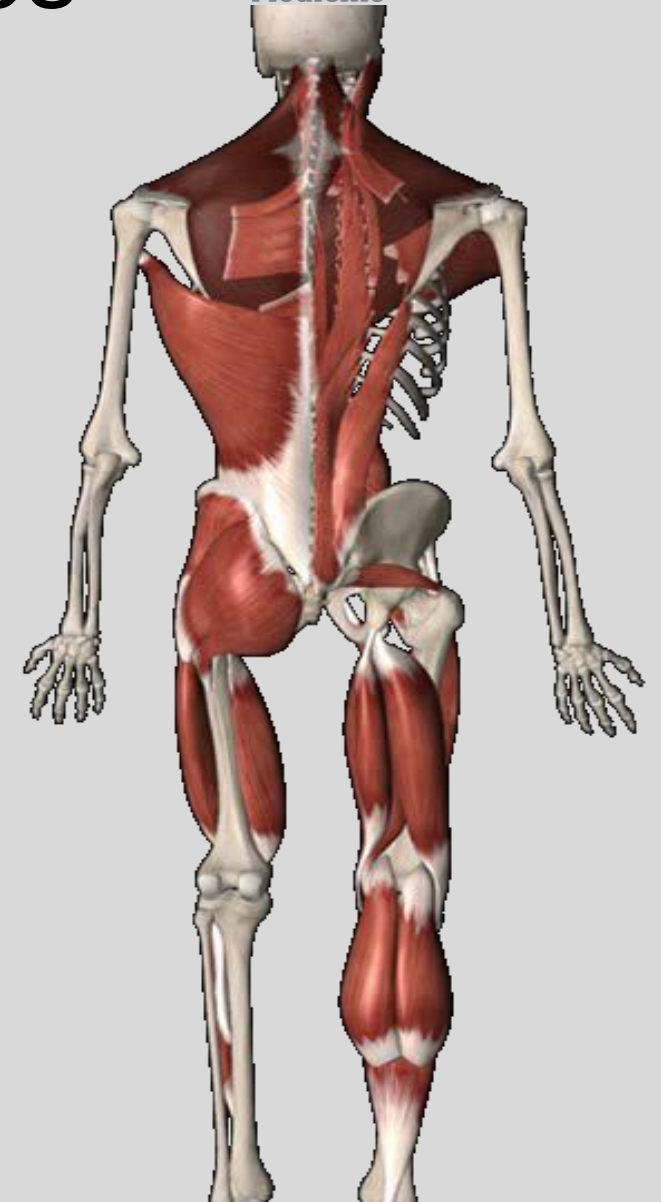


Table 2 – Posterior Muscles involved in Upper and Lower Crossed Syndromes

memorize

Posterior Postural Muscles	Posterior Phasic Muscles
Upper Cross	
Upper Trapezius	Mid-lower Trapezius
Levator Scapulae	Rhomboids
Cervical erector spinae	Latissimus dorsi
Lower Cross	
Quadratus lumborum	Gluteus minimus
Thoracolumbar erector Spinae	Gluteus medius
Hamstrings	Gluteus maximus
Piriformis	
Gastroc-soleus complex	



Clinical Association: PRONATION OF THE SUBTALAR JOINT and Lower Crossed Syndrome

- PRONATION of the subtalar joint and LOWER CROSSED SYNDROME
- Mechanical stress at the knee (genu valgus/valgum), hip and lumbar spine
- **Muscle imbalance** - tight/facilitated type I muscles vs. weak/inhibited type II muscles – KEY muscle (often) → GLUTEUS medius weakness

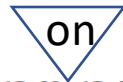
Clinical Correlation

Original Article

JER

<https://doi.org/10.12965/jer.1836606.303>

Journal of Exercise Rehabilitation 2019;15(1):123-128



The effect of a Janda-based stretching program range of motion, muscular strength, and pain in middle-aged women with self-reported muscular skeletal symptoms

Ji-Eun Kim, Tae-Beom Seo, Young-Pyo Kim*

Department of Kinesiology, College of Natural Science, Jeju National University, Jeju, Korea

The purpose of this study was to investigate the effect of a stretching program, based on the Janda approach and compared with a static stretching program, on range of motion (ROM), muscle strength and pain. Subjects for this study were 11 middle-aged women with self-reported muscular skeletal symptoms. The subjects were randomly divided into six women as an experimental group (stretching based on Janda approach group, SJAG) and five women as the comparison group (traditional static stretching group, TSSG). Static stretching program in both groups consisted of 14 types of exercise and strength program in SJAG was comprised of five isometric contractions. The stretching program was performed 3 times per week for 6 weeks. The ROM and muscle strength of the neck and shoulder region were measured and a numerical rating scale was used for measuring pain. Significant differences between groups were determined with two-way repeated analysis of variance and paired *t*-test. As a result of this study, the ROM did not show generally significant changes but was affirmatively improved in both groups. SJAG showed significant improvement in muscle strength and pain reduction compared to the TSSG. Consequently, a stretching program based on the Janda approach would be more effective on self-reported muscular skeletal symptoms caused by posture misalignment than a static stretching program.

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Keywords: Janda program, Range of motion, Muscular strength, Pain

Table 5. The result for the numeric rating scale (NRS)

Variable	Group	Pre	Post	F-value
NRS	TSSG	5.60±1.34	4.20±0.45	G×P: 20.928 [†]
	SJAG	6.83±1.17	1.83±0.75*	

Values are presented as mean±standard deviation.

TSSG, traditional static stretching group; SJAG, stretching based on Janda approach group; G×P, group×period.

**P*<0.05, significantly different from pretest. [†]*P*<0.05, significantly interaction among group and period.

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<http://www.e-jer.org>

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Kim, J., et. al., The effect of a Janda-based stretching program (on) range of motion, muscular strength, and pain in middle-aged women with self-reported muscular skeletal symptoms, Journal of Exercise Rehabilitation, 2019; 15(1):123-128, PMID: [30899747](https://pubmed.ncbi.nlm.nih.gov/30899747/); LINK: <https://pubmed.ncbi.nlm.nih.gov/30899747/>

Other Postural Considerations

- **Postural Compensation as an “Integrated Homeostatic Process”:**

- Begins in utero, continues throughout growth.
- The CNS and peripheral nervous system together learn to balance the postural homeostatic mechanisms.
 - “Each person will progressively compensate in a different way depending, in part, on his or her unique intrinsic and extrinsic risk factors”
 - “Postural preferences and patterns will conform within the resultant connective tissue structure”
 - “Neuromusculoskeletal structures will remodel to reflect the functional demands placed upon them”
 - “Functional capability will be limited by less than ideal structure.”

Other Postural Considerations

- **Tensegrity Model:**

- From Architect Buckminster Fuller (1929) →
Tension + Integrity = Tensegrity; **this can also apply to biological systems**, in addition to architecture.
- “In this model, it is recognized that:”
 - “Areas of hypomobile somatic dysfunction may cause compensatory hypermobility elsewhere”
 - “Hypermobility in one or more regions may affect compensatory hypomobility elsewhere”
 - “Fascial patterning and musculoligamentous tensions take on primary importance as the model’s connecting elements”

Other Postural Considerations

- **Coronal Plane:** Groups Curves
- **Sagittal Plane:** Kyphoses and Lordoses
- **Transverse Plane:** Rotational preferences and Transitional Regions (OA, CT, TL, LS)
- **Correlate with:** Somatic Dysfunction

Other Postural Considerations

- **Gravitational Strain Pathophysiology (GSP)**

signs & symptoms:

- Changes/alterations in postural alignment
- Recurrent somatic dysfunctions
- Lower back pain
- Referred pain to extremities
- Headache
- Somatovisceral effects/presentations
- Fatigue
- Weakness

- Progression to: **Postural Decompensation**, “when an individual’s functional homeostatic mechanisms are overwhelmed or when the degree of pathological change becomes structurally incapable of resisting gravitational force.”

Seffinger, Michael A., D.O., executive editor, **Foundations of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29**: “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera (*read under the Heading of “Common Postural Clinical Conditions,” and search for “Gravitational Strain Pathophysiology and Postural Decompensation”*)
LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?bookId=2582§ionId=209547150&resultClick=1#209547196>

Other Postural Considerations

- **Gravitational Strain Pathophysiology (GSP) Considerations:**
 - Intrinsic Factors
 - Extrinsic Factors
 - Tissue Responses:
 - Skeletal-Arthrodial (Wolff Law: “bone remodels over time in response to the stresses placed upon it.” – *Foundations*, 3rd Ed. p. 446)
 - Regional (Transition Zone) Fascial Response (J. Gordon Zink, D.O. and the Respiratory-Circulatory Model)
 - Muscular Response:
 - TriggerPoints (may be related to myotomal or sclerotomal pain patterns)
 - Tight vs. Weak muscle patterns, such as **the Upper Crossed and Lower (Pelvic) Crossed Syndromes**
 - Ligamentous Response
 - Response of Related Neural, Vascular Lymphatic and Visceral Elements

Seffinger, Michael A., D.O., executive editor, **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29:** “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera (*read under the Heading of “Common Postural Clinical Conditions,” and search for “Gravitational Strain Pathophysiology and Postural Decomensation”*)
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Other Postural Considerations

● Gravitational Strain Pathophysiology (GSP) Considerations:

from General Diagnostic Considerations:

- “The **purpose of postural diagnosis** therefore is to determine where in the continuum a given patient is and whether **the osteopathic postural model** could or should be integrated into that patient’s healthcare regimen. To this end, **the diagnostic process** should answer at least **four specific questions:**”
- (1) “Does the patient’s posture play a role in a patient’s given complaint?”
- (2) “Does the patient’s posture detract or limit the patient’s ability for self-healing?”
- (3) “Does the patient’s posture present a significant risk factor for pain, dysfunction, or pathology in the future?”
- (4) “Is the patient able to respond positively to a postural treatment regimen in a time- and cost-effective manner?”

Seffinger, Michael A., D.O., executive editor, **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29:** “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera (*read under the Heading of “Common Postural Clinical Conditions,” and search for “Gravitational Strain Pathophysiology and Postural Decomensation”*)
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Other Postural Considerations

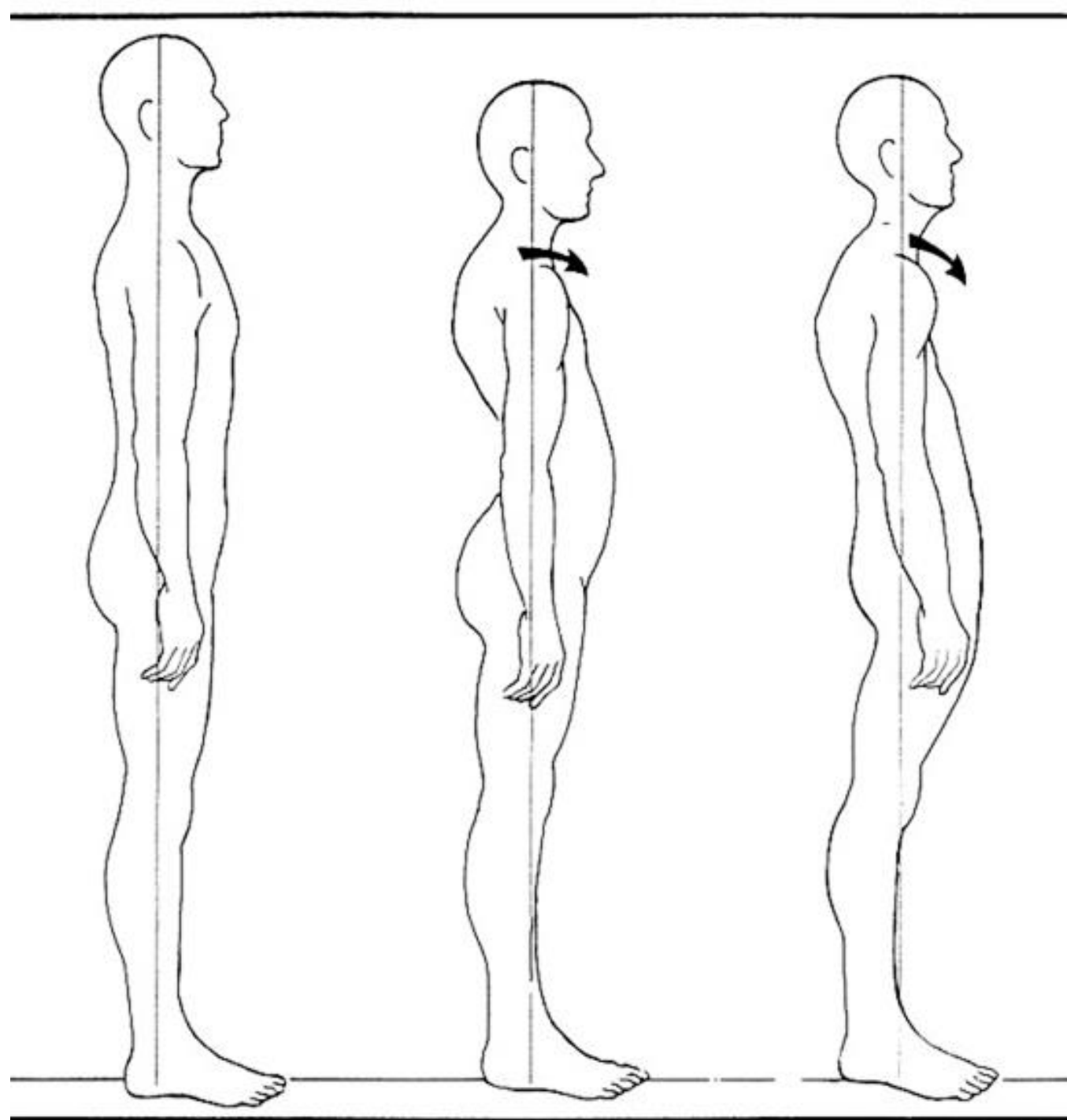
- **Gravitational Strain Pathophysiology (GSP)**
Considerations: from General Diagnostic Considerations:
 - “The **postural diagnostic approach** to answering these questions **relies heavily on observation and palpation**. It attempts to **correlate** other findings from the patient’s history and physical examination in order to determine if any pertinent signs or symptoms exist. It may also bring in **radiological or biomechanical tests** to help **quantify** the extent and type of postural change.”

Postural Considerations: General Treatment Strategies

- “(1) Postural Education and Compliance
- (2) OMT generally and to transition zones
- (3) Postural exercise
- (4) Static postural bracing
- (5) Functional postural orthotics
- (6) Other miscellaneous approaches”

- Symptoms-compensatory or facilitative mechanisms may cause:

- Cervical, thoracic, lumbosacral pain
- GI disorders, abdominal, pelvic and inguinal pain
- Headache, neck pain



General Treatment Strategies, cont.

Overall, there are many approaches that “are used to assist the body’s postural response to gravity. An **individually designed** conservative program includes **a carefully selected combination of patient education, OMT, exercise, and functional orthotics**. All of these are aimed at modifying the structure-function relationship and enhancing the body’s ability to self-heal. **Therefore, postural balancing requires an understanding of the biomechanical nature and functional anatomy of each patient and a full understanding of osteopathic philosophy.**

- “Patients are also awakening to the fact that current conventional medical interventions are often dangerous, costly, and ineffective.”
- “The U.S. medical system continues to be the third largest cause of death, after cardiovascular illness and cancer.”
- “To change this long-term trend, it is vital to employ therapeutics that are effective, efficient, and relatively less expensive and produce the least side effects.”
- “The profession needs to identify and treat the root cause of illness, be it of the body, mind, spirit, and/or the energy fields that support them.
- DOs need to develop, refine, and use their osteopathic skills so that they can adhere to Dr. Still’s reminder, ‘To find health should be the object of the doctor. Anyone can find disease.’”

Lower Crossed Syndromes: Exercises

**Patricia Kooyman, DO, Michael Terzella, DO and
Joanne Donoghue, PhD**

All of the following muscles play a role in the lower crossed syndrome.

Lower
Crossed Syndrome
Stretching

6

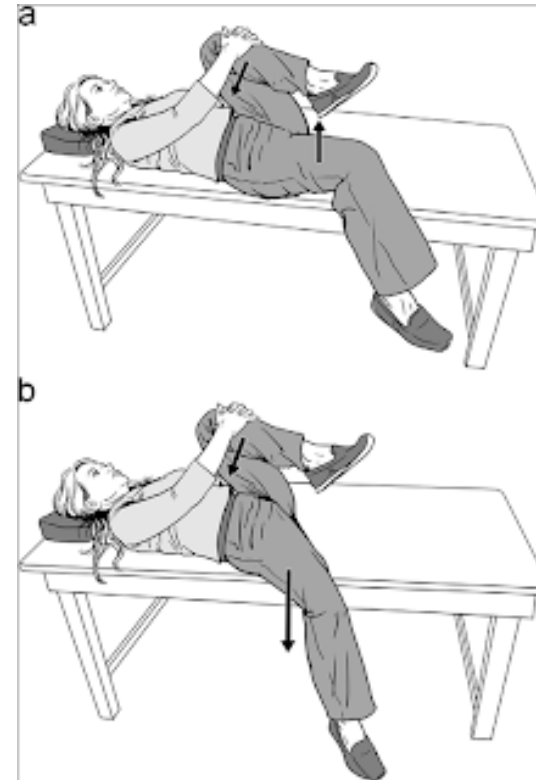
Iliopsoas Hip Flexor Stretch - Kneeling



Kneel and place opposite foot significantly forward
Hold on to couch or bed and move your pelvis forward while keeping your shoulders back
Feel stretch in front of hip of leg that is on the ground
Hold 30-60 seconds

7

Iliopsoas Hip Flexor Stretch – on back



Lie on bed on back. Pull one knee up towards chest.
Other foot stays hanging off the bench.
Hold for 30-60 seconds. Repeat on other side.
Feel stretch in front of hip of leg that is hanging.

8. Rectus Femoris stretch



Hold on to something, pull heel up toward buttocks. Feel stretch on front thigh. Hold for 30-60 seconds. Repeat on other side.



Eyes and head forward, slight bend in standing leg, toes resting on edge of chair

9. Adductor brevis Groin stretch



Stand or hold on to solid object.
Spread legs wide and move hips
forward lean to one side
Feel stretch in groin of straight leg
Hold 30-60 seconds

10

**Quadratus lumborum
Lateral Flexion Trunk
Stretch-sitting**



Sit on couch or bed, keep feet planted on ground and lean to the right as in picture. Hold 2 minutes.

11

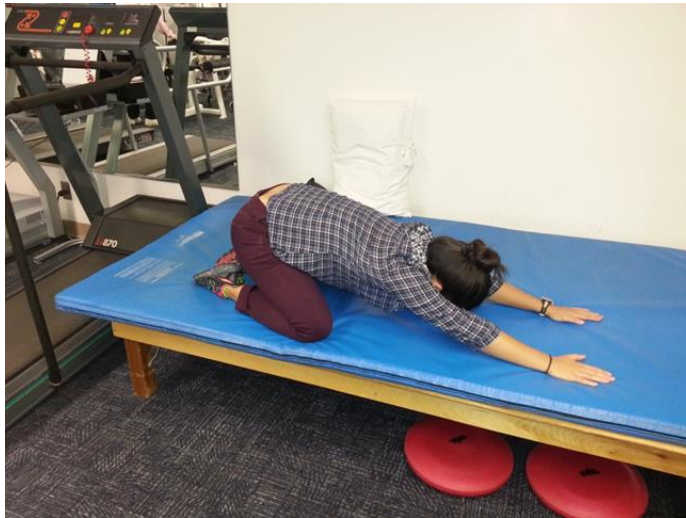
**Quadratus lumborum
Lateral Flexion Trunk stretch
Standing**



Do not lean forward.
Reach arms above the head and lean to one side. Change sides.
Hold 30-60 seconds

12

Erector Spinae Stretch Trunk Flexion



Lie prone. Reach forward and bring buttocks to the heels. Lower head to stretch cervical erector spinae
Hold 30-60 seconds. Repeat.

13

Erector Spinae Stretch Knees to chest



Lie supine. Bring knees to chest and lift head up (not in picture).
Hold 30-60 seconds. Repeat.

14

Hamstring Stretch



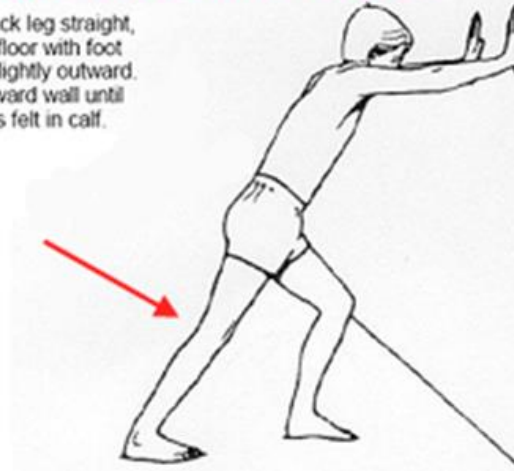
Sit on edge of solid chair
One leg is straight, the
other bent
Reach toward knee and
beyond if possible of
straight leg
Feel stretch in back of
straight leg
Hold 45 seconds

15

Gastroc-soleus complex stretch

Straight Leg Stretch for Gastroc

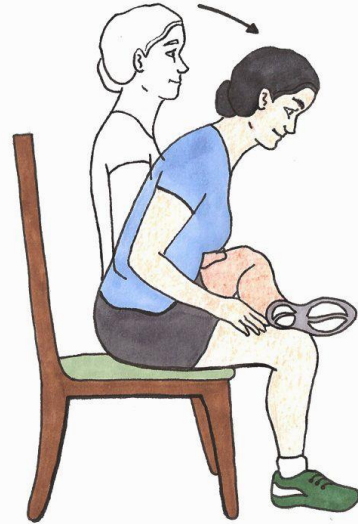
Keep back leg straight,
heel on floor with foot
turned slightly outward.
Lean toward wall until
stretch is felt in calf.



Hold on to solid object
such as kitchen
counter/wall
Place balls of feet on
2x4 or something a
little higher
Make sure heels are
on ground and knees
are straight
Feel stretch in calf
Hold 1-2 minutes

16

Piriformis stretch



Sitting in a chair, cross the affected side ankle onto the thigh. Turn head into the direction of the crossed leg. Hold 30-60 seconds

17

Supine piriformis stretch

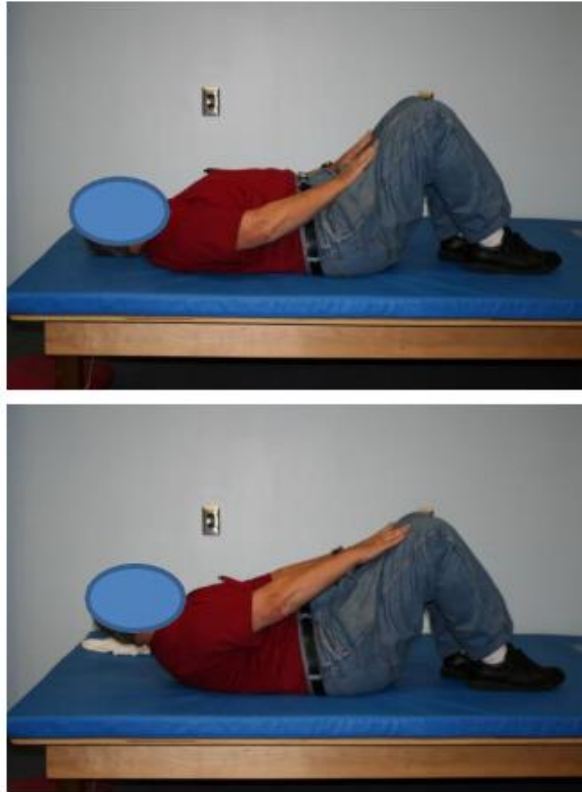


Lie supine, cross the affected side ankle onto the thigh. To increase stretch pull the opposite knee to chest. Hold 30-60 seconds

Lower
Crossed Syndrome
Strengthening

19

Upper Abdominals



Lie on back, knees bent so feet are flat on floor.
Stare at a spot on the ceiling and keep your eyes fixed on this spot as you reach toward the top of your knees.
Exhale as you come up, hold for one second and come back down.
Repeat 8-15 times (x 2 sets)

20

Lower Abdominals

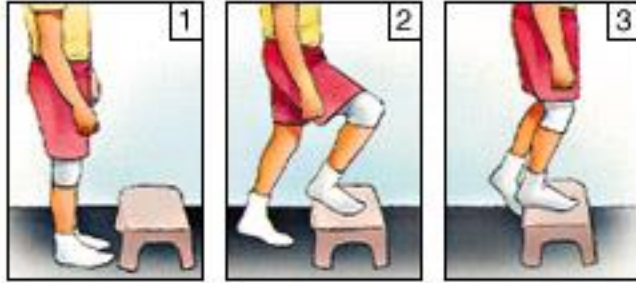


<https://www.mayoclinic.org/pelvic-tilt-exercise/img->

Lie on back, knees bent so feet are flat on floor.
Tighten your abdominal muscles by flattening your back on the floor and roll your pelvis as pictured above
Exhale as you come up, hold for one second and come back down.
Repeat 8-15 times (x 2 sets)

Vastus Intermedius, Vastus Lateralis, Vastus Medialis

21

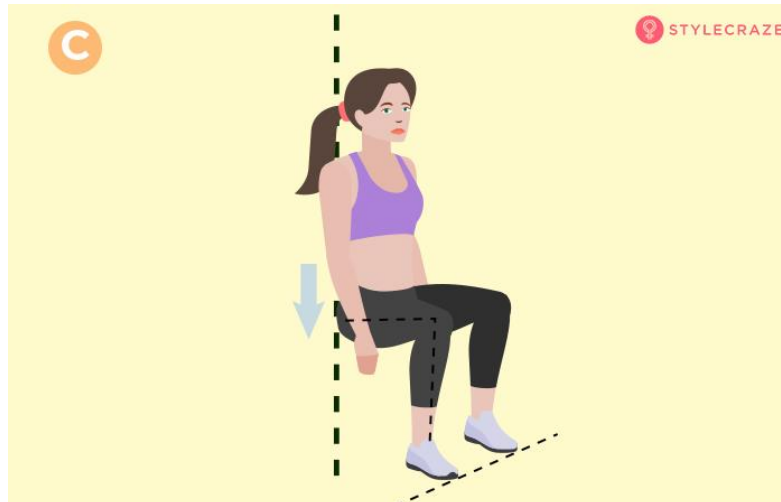


Step up onto a bench straightening the knee that you step up with. Repeat 8-15x.

22



Stand up on a step. **Hold on to something for balance.** Lower a heel onto the floor but do not put weight on it. Just touch and come back up to starting position. Repeat 8-15 times.

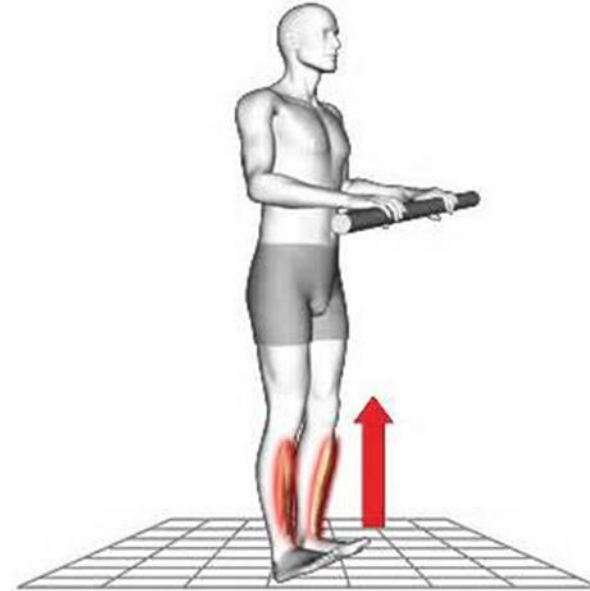


Keep back flat against wall, slide down until legs are at 90 degrees. Shift weight to the heels



23

Tibialis Anterior Toe Raises



Hold on to a solid object such
as kitchen counter
Raise up onto heels as high as
you can. Repeat 8-20x

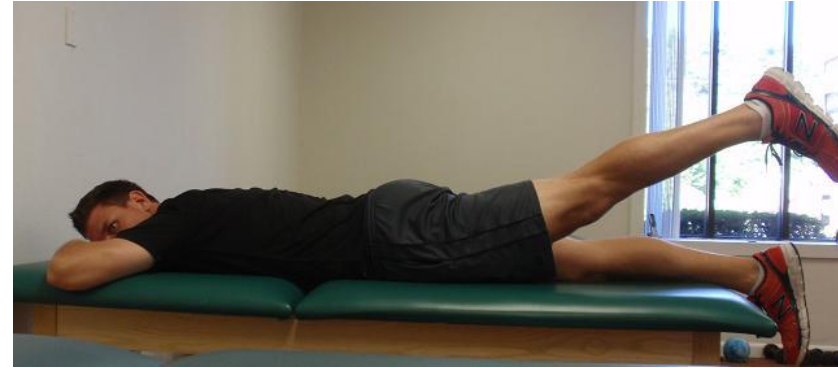
Gluteus maximus
Hip extension standing

27



Hold on to kitchen counter.
Keep knee straight and bring
leg back as far as you can
without leaning forward.
Repeat 12-15 times on each
side.

Hip extension on stomach



Lie on stomach. Keep knee straight, lift as high as you can.
Repeat 12-15 times on each side.

Gluteus medius, minimus
Hip abduction

28



Lie on side, abduct hip. Keep hand in front to prevent rolling back. Repeat 8-15x.

Gluteus medius, minimus
Hip Hike/ Stabilizing

29



Stand on a step, keep the standing leg straight and locked. Raise the opposite hip up. Repeat 8-15x.

Summary Slide

- The Lower (Pelvic) Crossed Syndrome provides a clear framework from which to view a patient's postural compensations and subsequently target treatment approaches (*which can include an Exercise Prescription, as noted in the examples*).
- Other postural considerations are manifold and important to consider in a patient's clinical presentation.
- Application of the osteopathic perspective and the osteopathic philosophy can be beneficial for a patient's overall well being.

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1. Seffinger, Michael A., D.O., executive editor, **Foundations** of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, **Fourth Edition**, Wolters Kluwer, 2018, **Chapter 29**: “Postural Considerations in Osteopathic Diagnosis and Treatment” by Michael L. Kuchera (*read under the Heading of “Muscular Response” and see Table 29.3 Examples of Postural (Hypertonic-Prone) Muscles and Symptoms, and Table 29.4 Examples of Phasic (Pseudoparesis-Prone) Muscles and Symptoms*, **LINK**: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547285>
 - *And under the Heading of “Summary” see the “Clinical Pearls” Box Section as relates to Tables 29.3 and Table 29.4 as noted above, LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547738>*
 - *And read from Heading of “Common Postural Clinical Conditions” (especially including the section on “Gravitational Strain Pathophysiology and Postural Decompensation”) to just before the Heading of “General Diagnostic Considerations”), LINK: <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582#209547196>*

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- **Chapter 61:** “Dynamic Strain and the Bioenergetics Model” by JAN T. HENDRYX, **LINK:** <https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261507611&bookid=3456>
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8. Kim, J., et. al., The effect of a Janda-based stretching program (on) range of motion, muscular strength, and pain in middle-aged women with self-reported muscular skeletal symptoms, Journal of Exercise Rehabilitation, 2019; 15(1):123-128, PMID: [30899747](https://pubmed.ncbi.nlm.nih.gov/30899747/); LINK: <https://pubmed.ncbi.nlm.nih.gov/30899747/>

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